$$\frac{\mathrm{d}}{\mathrm{d}t}\langle \hat{P}^2 \rangle_t = -M(\Omega^2 + \widetilde{\Omega}^2)\langle \hat{X}\hat{P} + \hat{P}\hat{X} \rangle_t - 4\gamma \langle \hat{P}^2 \rangle_t + 2D. \tag{5.37}$$